# MALWANDLA HLONGWANE

## Junior Android Developer | Mobile App Developer

Pretoria, Gauteng, South Africa | 065 883 1240 | MalwandlaHlongwaneICT@outlook.com
**Portfolio:** malwandla21-ict.github.io/MH-Profile | **GitHub:** github.com/Malwandla21-ict
**LinkedIn:** linkedin.com/in/malwandla-hlongwane-8484603a3

## PROFESSIONAL SUMMARY

Final-year Diploma in ICT (Applications Development) student at the University of Mpumalanga who builds mobile apps in Flutter and native Android (Java). Built and launched ZAKA, a Flutter and Supabase savings app for Android and web, and HealthWise, an AI symptom checker using Room, Retrofit, AlarmManager and Material 3; currently building STAMS, a Flutter app backed by a Spring Boot REST API. Strong Java and backend foundation from Jakarta EE, JavaFX and Node.js projects, covering both the app and the API behind it. Seeking a Junior Android Developer, Mobile Developer or internship role.

## TECHNICAL SKILLS

- **Languages:** Java, Dart, SQL, XML; Kotlin (coursework level)
- **Android:** Android Studio, Android SDK, Activities and lifecycle, XML layouts, Material Components 3, RecyclerView, intents, notifications, AlarmManager, BroadcastReceiver
- **Data & Networking:** Room (SQLite) with migrations, SharedPreferences, Retrofit, Gson, REST APIs, JSON, AI API integration (Hugging Face)
- **Flutter:** Dart, Riverpod, go_router, local notifications, secure storage, widget and unit tests, signed release APKs
- **Backend:** Supabase (PostgreSQL, Auth, Storage), Spring Boot, Spring Security (JWT), Node.js, Express.js, MySQL
- **Architecture & Tools:** Repository pattern, MVC/MVVM, object-oriented design, Git, GitHub, Postman, debugging and testing
- **Also covered in coursework:** Firebase, Google Maps API and location services, multimedia, signed APK/AAB and Play Store deployment
- **Professional:** Problem solving, user-focused design, teamwork, technical documentation, fast learner

## PROJECT EXPERIENCE

### ZAKA | Savings Tracker & Spitori Dictionary

*Cross-platform app for Android and web | Deployed live | Private repository, code available on request*
**Tech:** Flutter, Dart, Riverpod, go_router, Supabase (PostgreSQL, Auth, Storage, Edge Functions), pgTAP, Cloudflare Pages

- Built and launched a savings app inspired by Pretoria's Spitori culture: savings goals with daily, weekly or monthly plans, ahead/behind tracking, payday reminders, charts, and a free Spitori dictionary.
- Organised the app by feature into data, domain and presentation layers, using Riverpod for state, with only the data layer allowed to call Supabase.
- Wrote 435 Flutter tests and 198 pgTAP database checks covering row-level security, balance rules, limits and admin access.
- Scheduled Android reminders with local notifications on South African time, kept the login session in secure storage, and shipped signed release APKs split by phone type.

**Live site:** zaka-3hi.pages.dev

### HealthWise | AI Symptom Checker Android App

*Native Android application*
**Tech:** Java, Android SDK, XML, Material Components 3, Room (SQLite), Retrofit, Gson, AlarmManager, Hugging Face API

- Built a native Android app with a four-step guided symptom check that ranks likely conditions with confidence scores and gives urgency advice.
- Integrated an AI chat assistant through the Hugging Face Inference API using Retrofit, keeping conversation context and suggesting follow-up questions.
- Stored check history, health profile and medication data in Room, with schema migrations that keep user data across upgrades.

- Built pill reminders with AlarmManager, BroadcastReceiver and notifications that work when the app is closed, with a mark-as-taken action.
- Used a repository pattern to separate Activities (UI) from data access and scoring logic.
- Designed light and dark themes with Material 3, voice input for symptoms and adaptive icons; supports Android 7.0 (API 24) and up.

**GitHub:** github.com/Malwandla21-ict/HealthWise

### STAMS | Student Accommodation Management System

*Mobile app + REST API | In progress*

**Tech:** Java 21, Spring Boot, Spring Security, JWT, Spring Data JPA, Hibernate, PostgreSQL, Flutter, Dart, Postman

- Building a Flutter mobile app and Spring Boot REST API for landlords to manage properties, rooms and student residents.
- Designing a maintenance workflow where students report issues with photos, track them from Reported to Resolved, and nudge landlords when an issue stalls.
- Implementing JWT authentication with Spring Security and testing API endpoints in Postman.
- Next phases: Swagger/OpenAPI docs, unit tests, Docker and GitHub Actions CI/CD.

**GitHub:** github.com/Malwandla21-ict/STAMS-Student-Accomadation-Management-System-

## ADDITIONAL PROJECTS

### Taskify | Campus Task & Equipment Marketplace

*Full-stack web application | Deployed live*

**Tech:** Node.js, Express, MySQL, JavaScript, HTML, CSS, REST API, JWT, Render

- Built and deployed a full-stack marketplace where students post tasks, hire services, buy items and rent equipment.
- Secured the platform with JWT authentication, TOTP two-factor sign-in for admins, bcrypt password hashing, Helmet security headers, rate limiting and server-side input validation.
- Wrote 134 automated tests across 13 test suites (Node.js test runner) covering rentals, sales, disputes and tasks, all passing.

**GitHub:** github.com/Malwandla21-ict/taskify | **Live site:** taskify-marketplace.co.za

### FUEL SA | Fuel Ordering & Station Management System

*Java desktop application*

**Tech:** Java 17, JavaFX 21, FXML, CSS, SQLite (JDBC), Maven, BulkSMS REST API

- Built a desktop app where customers register vehicles and request fuel with a live cost estimate, and admins approve or reject requests from a dashboard.
- Stored passwords as salted PBKDF2 hashes and kept API credentials out of source code using environment variables.
- Organised code into controllers, DAOs, models and services, with a demo mode that loads sample data.

**GitHub:** github.com/Malwandla21-ict/FuelSA

## EDUCATION

### Diploma in Information and Communication Technology (Applications Development)

University of Mpumalanga | Expected completion: 2026 | Academic average: 75–78%

**Relevant coursework:**

- **Android Application Development:** app architecture and lifecycle, XML/Material Design UI, navigation and intents, SharedPreferences, SQLite/Room, location services, REST/Firebase integration, testing, debugging, Play Store deployment
- **Application Development II (Java):** OOP, GUI development, collections, recursion, searching and sorting algorithms, file I/O, JDBC
- **Software Engineering:** SDLC, requirements engineering, UML, design patterns, Agile/Scrum, software testing